Module 12

HashTable – Simple Implementation

Last time:

Knowing the index value of an array makes us to find an object in constant time (random access).

Thus, our savior to make search faster is a hash function that returns an index value of an object.

However, we also learned that collisions are a fact of life.

We looked into a few ways to deal with collisions.

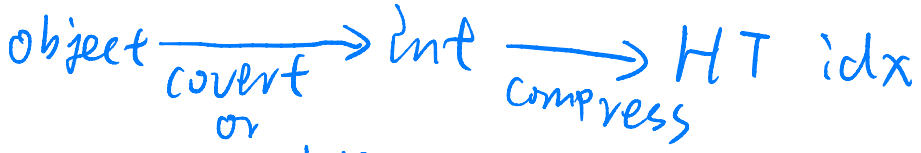
- **Linear Probing**: When there is a collision, we try to find an empty cell sequentially and put the value into the nearest empty cell. However, this approach has an issue of forming clusters (primary clustering) and the performance can get bad. It is necessary to rehash to keep the load factor from time to time.
- There is also **Quadratic Probing** that has another subtle clustering issue called secondary clustering due to the fixed interval of probing.
- To solve this clustering issue, there is another workaround, called **Double Hashing**, which uses two hash functions. One is to calculate the hash value and the other is to decide the step size of probing.
- **Separate Chaining**: The other workaround is to have a linked list at each index. This allows us to not to worry too much about load factor. However, we do not want to make the linked lists become too full either.

Time to get our hands dirty by implementing a version of hash table!

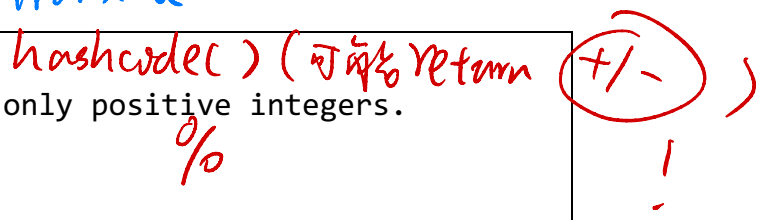
We will implement a version of hash table using ***linear probing as its collision resolution method*.**

Major operations of our hash table are:

- search(int key)
- delete(int key)
- insert(int key)



HashTableInterface interface




```
/**
 * HashTable interface that takes only positive integers.
 * No mapping. Just keys
 */
public interface HashTableInterface {
    /**
     * Returns true when the key is found.
     * @param key int key to search
     * @return boolean true if found, false not found
     */
    boolean search(int key);

    /**
     * Deletes and returns an int key from the table.
     * @param key int key to delete
     * @return the deleted int key (if not found, -1)
     */
    int delete(int key);

    /**
     * Inserts an int key to the table.
     * @param key int key to insert
     */
    void insert(int key);
}
```

HashTable class

```
public class HashTable implements HashTableInterface {
    private static final DataItem DELETED = new DataItem(-1);
    private DataItem[] hashArray;

    // precondition: initialCapacity is a positive int
    public HashTable(int initialCapacity) {
        hashArray = new DataItem[initialCapacity];
    }

    // static nested class
    private static class DataItem {
        private int key;

        DataItem(int k) {
            key = k;
        }
    }

    // TODO implement methods here
}
```

First things first: Hashing method

```
// private helper method for hashing a key value
private int hashFunc(int key) {
    return key % hashArray.length;
}
```

Looks too simple? Well… for now.

Searching for a key

```
@Override
public boolean search(int key) {
    int hashVal = hashFunc(__key__);
    while (hashArray[hashVal] != __null__) {
        if (hashArray[hashVal].__key__ == key) {
            return true; // found
        }
        __hashVal++__;
        // wrap around
        hashVal = __hashVal % hashArray.length__;
    }
    return false; // cannot find
}
```

Deleting a key (Practice for you!)

```
@Override
public int delete(int key) {
```

```java
@Override  0 个用法  新 *
public int delete(int key) {
    int hashVal = hashFunc(key);
    while (hashArray[hashVal] != null) {
        if (hashArray[hashVal].key == key) {
            int temp = hashArray[hashVal].key;
            hashArray[hashVal] = DELETED;
            return temp;
        }
        hashVal++;
        hashVal = hashVal % hashArray.length;
    }
    return -1;
}
```

Inserting a key

```
@Override
public void insert(int key) {
    int hashVal = hashFunc(key);

    while (hashArray[hashVal] != null &&
           hashArray[hashVal] != DELETED) {
        hashVal++;
        // wrap around
        hashVal = hashVal % hashArray.length;
    }
    DataItem item = new DataItem(key);
    hashArray[hashVal] = item;
}
```

A quick thought about open addressing:
When there are many deletions, open addressing may not be efficient. More specifically, search will take longer. Especially, unsuccessful searches could be quite slow.

We had a chance to see how a hash table can be implemented, at least, using linear probing as its collision resolution method. ***There is a lot of room for improvement in the code. It is just enough to show the basic concepts.***

Moving on, our next question is how we can implement our own hash algorithm? In other words, what makes a good hash function?

1. Quick to compute.
2. Evenly + randomly distribute keys in HT.

What makes a good hash function?

Quick computation is the key to a good hash function. Thus, a hash function with many multiplications and divisions is NOT a good idea.

The purpose of a hash function is to take a range of key values and transform them into index values in a way that the ***key values are distributed randomly across all the indices*** of the hash table.

Key values may be completely random or not so random.

1. Random key values

If the key values are random and positive, then we can simply find index values by the following simple operation just like our code before.

Object.hashCode() might be negative

index = key % hashArray.length;

2. Non-random key values

For example, there is a database that uses car-part numbers as key values.

033-400-03-94-05-0-535

This is interpreted as follows:

- Digits 0-2: Supplier number (1 to 999, currently up to 70)
- Digits 3-5: Category code (100, 150, 200, 250, up to 850)
- Digits 6-7: Month of introduction (1 to 12)
- Digits 8-9: Year of introduction (00 to 99)
- Digits 10-11: Serial number (1 to 99, never exceeds 100)
- Digits 12: Toxic risk flag (0 or 1)
- Digits 13-15: Checksum (sum of the other fields)

Based on the interpretations provided, the key value should be 0,334,000,394,050,535 for the particular part number shown above.

However, we can say that there is no guarantee that we will have random numbers between 0 to 9,999,999,999,999,999.

Some work should be done to have these part numbers to form a range of more random numbers.

- **Don't Use Non-Data**: The key values should be squeezed as much as it could. For example, category code should be changed to be from 0 to 15. Also, the checksum should be removed because it is a derived number from other information and does not add any new information.
- **Use All the Data**: Other than the non-data values, we need to use all the data values. Don't just use the first four digits, etc.
- **Use a Prime Number for the Modulo Base**: Which means the table length should be a prime number. For example, if the table array length is 50, then all the multiples of 50 in our car-part numbers will be hashed into the same index.
- **Use Folding**: Another reasonable hash function involves breaking keys into groups of digits and adding the groups.

  SSN example: 123-45-6789

  In case table length is 1009: Break the number into three groups of three digits. (123+456+789 = 1368 % 1009 = 359).

  In case table array length is 101: Break the number into four two-digit numbers and one one-digit number. (12+34+56+78+9 = 189 % 101 = 88).

  This way you can *distribute* the numbers better.

To sum it up, it is critical for you to **examine your key values carefully** and implement hash function to *remove any irregularity in the distribution of the key values*. More on this in the next module!